

RACHANA KAFLE, B. Optometry

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EDUCATION

PhD, Physiological Optics and Vision Science

Aug 2024 – Present

University of Houston College of Optometry, Houston, TX | GPA: 3.93 / 4.00

- Dissertation: Multimodal Biomarker Development for Early Detection of Diabetic Ocular Complications
- Primary Mentor: Dr. Wendy W. Harrison, OD, PhD, FAAO
- Co-Mentors: Dr. Daniel R. Coates, PhD; Dr. Maria K. Walker, OD, PhD, FAAO

Bachelor of Optometry and Vision Sciences

Feb 2019 – Nov 2023

National Academy of Medical Sciences University, Kathmandu, Nepal | GPA: 3.95 / 4.00

- Thesis: Specular Microscopic Analysis of Corneal Endothelial Parameters in Keratoconus and Controls

RESEARCH EXPERIENCE

Graduate Research Assistant — Diabetes Biomarker Discovery Lab

Dec 2024 – Present

University of Houston College of Optometry | Mentor: Dr. Wendy W. Harrison; Co-Mentor: Drs. Daniel R. Coates & Maria K. Walker

- Design and conduct human-subject studies using TSLO, mfERG, OCT/OCTA, IVCN, and psychophysical testing to identify early markers of diabetic ocular dysfunction
- Quantify fixational eye movements from TSLO recordings including saccade amplitude, velocity, acceleration, drift, and fixation stability; identified directional asymmetry in T2DM
- Develop hybrid optic-nerve-head tracking algorithm (kernelized correlation filter + normalized cross-correlation) to passively extract fixation stability from routine mfERG fundus video
- Investigate corneal sub-basal nerve plexus (SBNP) morphology across controls, prediabetes, and T2DM using IVCN; pilot data (n=11) show HbA1c correlates negatively with corneal nerve branch density and fiber density
- Developing semi-automated pipeline (FIJI/ImageJ) for corneal nerve fiber thickness, beading size, and beading density — micro-structural markers that may detect subclinical neuropathy before standard parameters become abnormal
- Integrate IVCN with mfERG across 70 participants to determine whether corneal nerve degeneration and retinal electrophysiological dysfunction share a common disease trajectory; P1 implicit time correlates negatively with corneal nerve loss
- Contribute to longitudinal study tracking corneal nerve, tear protein, and anterior-segment changes in collaboration with a metabolic health laboratory at UH
- Contribute to preclinical study investigating dose-dependent effects of semaglutide (GLP-1 receptor agonist) on retinal VEGF expression in a high-fat-diet rat model using whole-mount immunostaining, qPCR, and quantitative image analysis
- Integrate optical imaging, electrophysiology, clinical measures, and quantitative image analysis across multimodal datasets using Python, MATLAB, SPSS, and JASP

Research Assistant — Nepal Eye Hospital

Jan 2023 – May 2023

National Academy of Medical Sciences University, Kathmandu, Nepal | Mentor: Dr. Raju Kaiti

- Conducted specular-microscopy analysis of corneal endothelial cell density, coefficient of variation, and hexagonality in more than 200 eyes with keratoconus and age-matched controls
- Performed clinical imaging and data analysis with applications to ophthalmic assessment, surgical planning, and specialty contact-lens management; manuscript in preparation

CLINICAL EXPERIENCE

Optometrist Intern — Nepal Eye Hospital, NAMS

May 2023 – Nov 2023

- Specialty elective rotation, Cornea and Contact Lens Department (Apr–Jun 2023): specialty contact lens fitting for irregular corneas including keratoconus and post-surgical cases
- Comprehensive eye examinations including primary diabetic retinopathy screening and staging
- Proficient in advanced corneal diagnostic imaging: IVCN, corneal topography, OCT, specular microscopy, ocular biometry, ERG, and slit-lamp biomicroscopy

PUBLICATIONS

Peer-Reviewed Journal Articles

- Published Kafle R**, Shyangbo R, Regmi A, Kaiti R. "Developmental Delay and Oculo-Visual Deficits: Challenges and Interventions — A Narrative Review." 2026.
- Published Kafle R**, Kaiti R, Shrestha P. "Pediatric Cataract and Biometry: Full of Challenges Yet Rewarding." *Optometric Vision Performance*. 2024.
- Published Kaiti R, Kafle R**, Shyangbo R, Dahal M, Maharjan S, Dhakal A. "Presbyopia and Recent Advances in its Management — A Review." *EC Ophthalmology*. 2020.
- Published Kafle R**, Owusu-Afriyie B, Wu C, Coates D, Harrison W. "Fixational Saccades and Drift Parameters are Altered in Poorly Controlled Type 2 Diabetes Mellitus: A Pilot Study." *Scientific Reports*. 2026.
- Under Review Kaiti R**, Dahal M, Shyangbo R, **Kafle R**, Sharma S. "Prevalence of Non-Strabismic Binocular Vision Anomalies in Asthenopia." 2026.
- In Prep Kafle R**, Shrestha P, Kaiti R. "Specular Microscopic Analysis of Corneal Endothelium in Keratoconus and Controls." 2026.

Peer-Reviewed Conference Presentations — First Author

- Accepted Kafle R**, Walker MK, Owusu-Afriyie B, Basak S, Harrison WW. "Corneal and Retinal Neural Degeneration May Occur Concurrently in Type 2 Diabetes Mellitus." AAO Annual Meeting. 2026.
- Paper Kafle R**, Owusu-Afriyie B, Wu CS, Coates DR, Harrison WW. "Saccade and Drift Parameters are Altered in Glucose Dysfunction." ARVO Annual Meeting, Denver, CO. 2026.
- Paper Kafle R**, Owusu-Afriyie B, Wu CS, Coates DR, Harrison WW. "Measurement of Fixation Stability in Type 2 Diabetes." AAO Annual Meeting, Boston, MA. 2025.

Co-Authored Conference Presentations

- Accepted Lehnert E**, Peng A, **Kafle R**, et al. "Evaluation of the Relationship of Nutrition with Retinal Structure and Function in Type 2 Diabetes." AAO Annual Meeting. 2026.
- Accepted Peng A**, Lehnert E, Owusu-Afriyie B, **Kafle R**, et al. "Early Lens Color Alterations Begin in Prediabetes and are Correlated with Retinal Function." AAO Annual Meeting. 2026.
- Accepted Owusu-Afriyie B, Kafle R**, Basak S, et al. "Oral Glucose Tolerance Test is Associated with Retinal Function and Structure in Prediabetes and Type 2 Diabetes." AAO Annual Meeting. 2026.
- Poster Awuni HA**, Owusu-Afriyie B, Venkatesh S, **Kafle R**, et al. "Retinal Arterioles and Venules are Remodeled in Type 2 Diabetes But Not Prediabetes." AAO Annual Meeting. 2026.
- Poster Basak S**, Owusu-Afriyie B, **Kafle R**, Harrison W, Redfern R. "Phenotypic Changes in Tear Film and Corneal Structure Across Control, Prediabetes, and Early Type 2 Diabetes." ARVO Annual Meeting. 2026.
- Poster Owusu-Afriyie B**, Kafle R, Basak S, et al. "Retinal and Choroidal Perfusion Density Are Impacted in Both Prediabetes and Type 2 Diabetes." ARVO Annual Meeting. 2026.
- Poster Owusu-Afriyie B**, Wu CS, Burhans L, **Kafle R**, et al. "Multimodal Characterization of Early Retinal Structural and Functional Changes in Prediabetes and Type 2 Diabetes." AAO Annual Meeting, Boston, MA. 2025.

AWARDS & FELLOWSHIPS

Joe and Janet Early Cornea Research Award <i>American Academy of Optometry</i>	2026
Graduate Student Achievement Award <i>University of Houston</i>	2026
Anh Lac Tran Award <i>Department of Psychology, University of Houston</i>	2026
Cullen Travel Fellowship	2026

University of Houston — Support for ARVO 2026, Denver, CO

AAO Travel Grant for ARVO 2026
American Academy of Optometry Foundation — Support for ARVO 2026

OCCRS Travel Grant 2025
Optometric Cornea, Cataract and Refractive Society, AAO Foundation — AAO 2025, Boston, MA

Presidential Fellowship 2024 – 2026
University of Houston — Competitive university-wide award for exceptional academic achievement

Merit-Based Full Tuition Scholarship 2024 – Present
University of Houston College of Optometry — Renewable annual award for outstanding academic achievement

TEACHING EXPERIENCE

Graduate Teaching Assistant Aug 2024 – Present
University of Houston College of Optometry

- Instruct first-year optometry and vision science students in Histology, Human General Anatomy, and Ocular Anatomy with emphasis on structure-function relationships
- Develop course materials, conduct lab sessions, and hold office hours

Clinical Teaching Assistant 2022 – 2023
Nepal Eye Hospital, National Academy of Medical Sciences University

- Supervised and instructed junior optometry students during clinical rotations in comprehensive eye care and diagnostic imaging

TECHNICAL SKILLS

Corneal Imaging and Anterior Segment

In Vivo Confocal Microscopy (IVCM): corneal nerve quantification including CNFD, CNBD, CNFL, fiber thickness, beading morphology; Specular Microscopy; Corneal Topography; Slit-Lamp Biomicroscopy; OCT/OCTA

Electrophysiology and Retinal Imaging

Multifocal ERG (mfERG); Tracking Scanning Laser Ophthalmoscope (TSLO) for fixational eye movement analysis; Fundus Photography; Retinal Immunostaining

Quantitative Image Analysis

Fiji/ImageJ (nerve tracing, morphometry, beading measurement); ACCMetrics; NeuronJ; Celleste; Adobe Suite

Programming and Statistics

Python (NumPy, SciPy, pandas, OpenCV, scikit-learn); MATLAB; SPSS; JASP; Multivariate modeling; Logistic regression; ROC curve analysis; Intraclass correlation; Mixed-effects models; Signal detection theory

Clinical

Specialty contact lens fitting (keratoconus, irregular corneas); Comprehensive ophthalmic examination; Diabetic retinopathy diagnosis and staging; qPCR; Whole-mount retinal immunostaining

PROFESSIONAL SERVICE AND MEMBERSHIPS

Moderator, Posterior Segment Poster Session 2025
AAO Annual Meeting, Boston, MA

Volunteer, Fellows Doing Research SIG 2025
AAO Annual Meeting, Boston, MA

Member-in-Training — ARVO 2025 – Present
Association for Research in Vision and Ophthalmology

Student Member — AAO 2024 – Present

American Academy of Optometry

Member — VSS

2026 – Present

Vision Sciences Society

Licensed Optometrist

2024

Nepal Health Professional Council

REFERENCES

Available upon request.